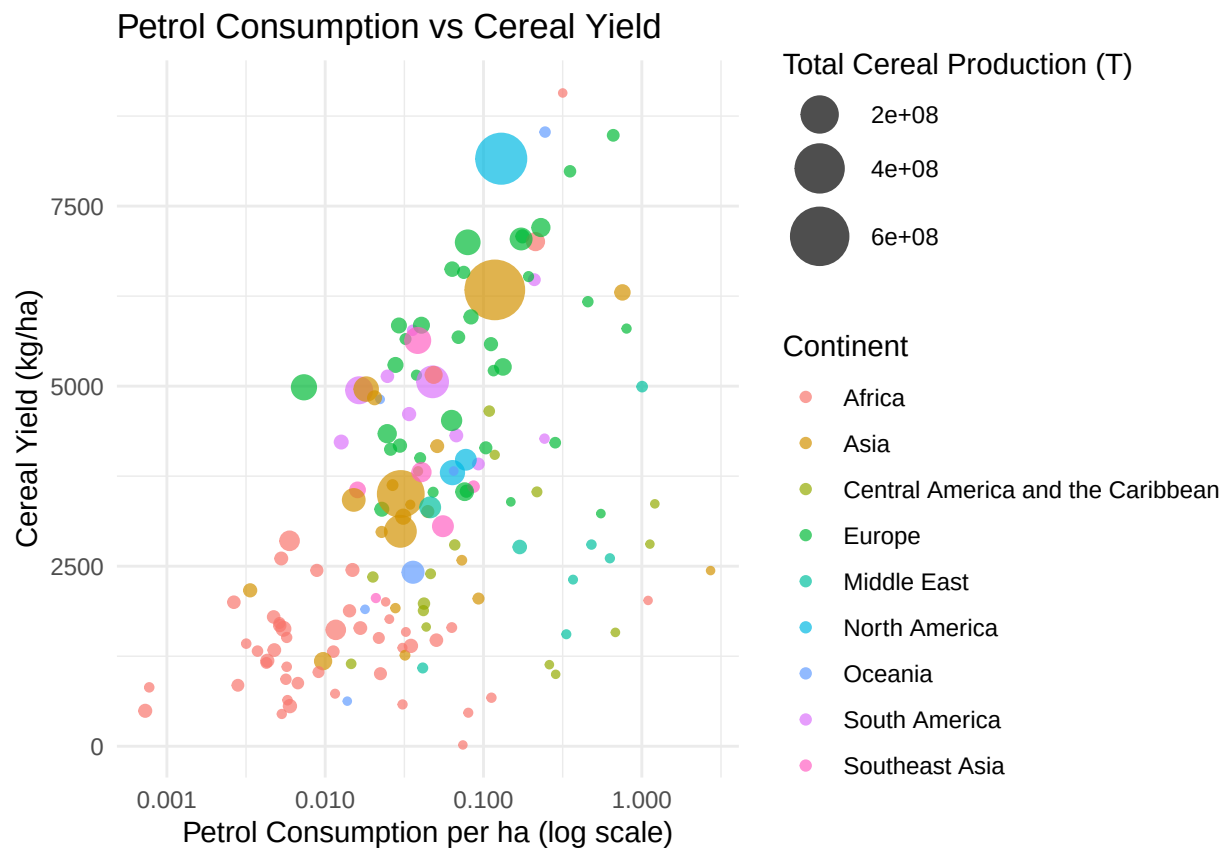


Question5

2025-06-18

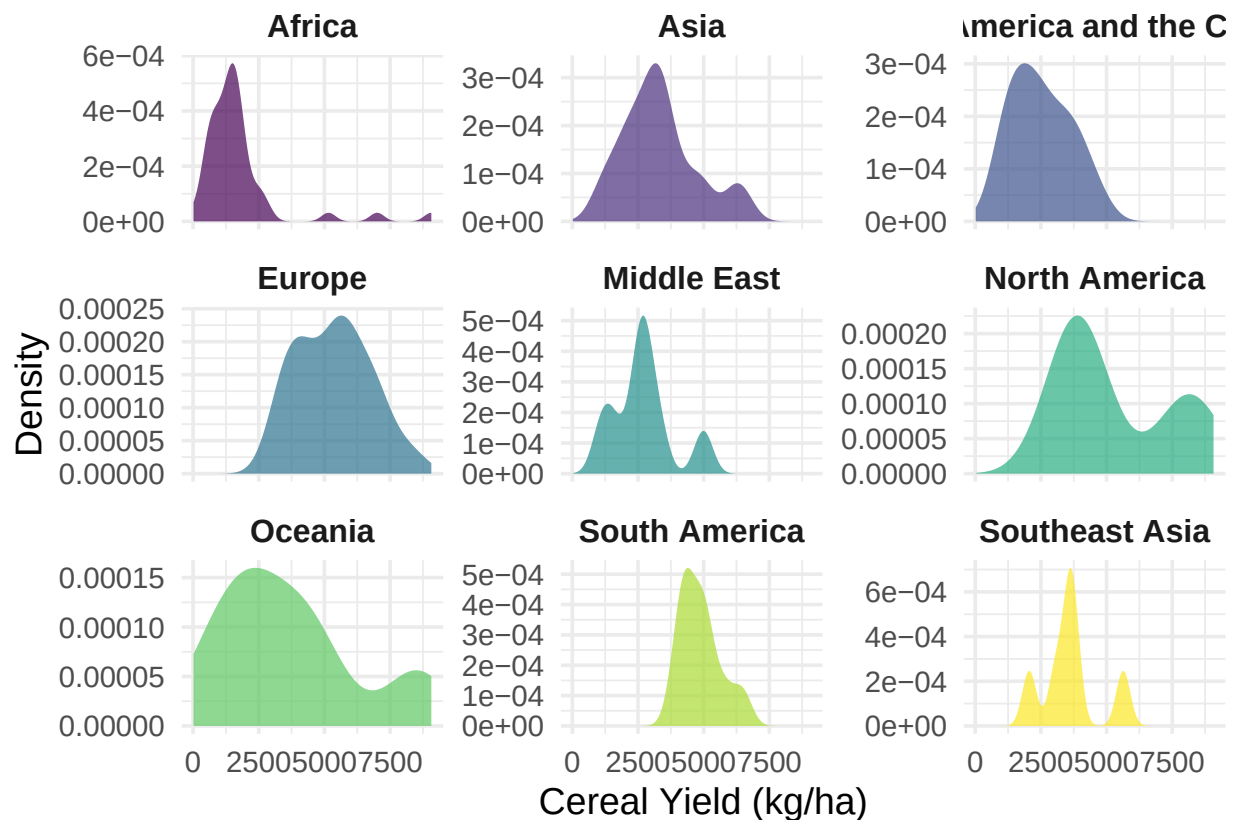
Can we identify global patterns in agricultural performance and energy-water usage among countries ?

overview



This scatter plot suggests a potential positive correlation between petrol consumption per hectare and cereal yield. Additionally, countries with higher total cereal production (indicated by the size of the points) also tend to exhibit higher petrol usage and yield levels.

A plausible hypothesis emerging from this visualization is that countries with higher energy inputs may achieve more productive agricultural systems.

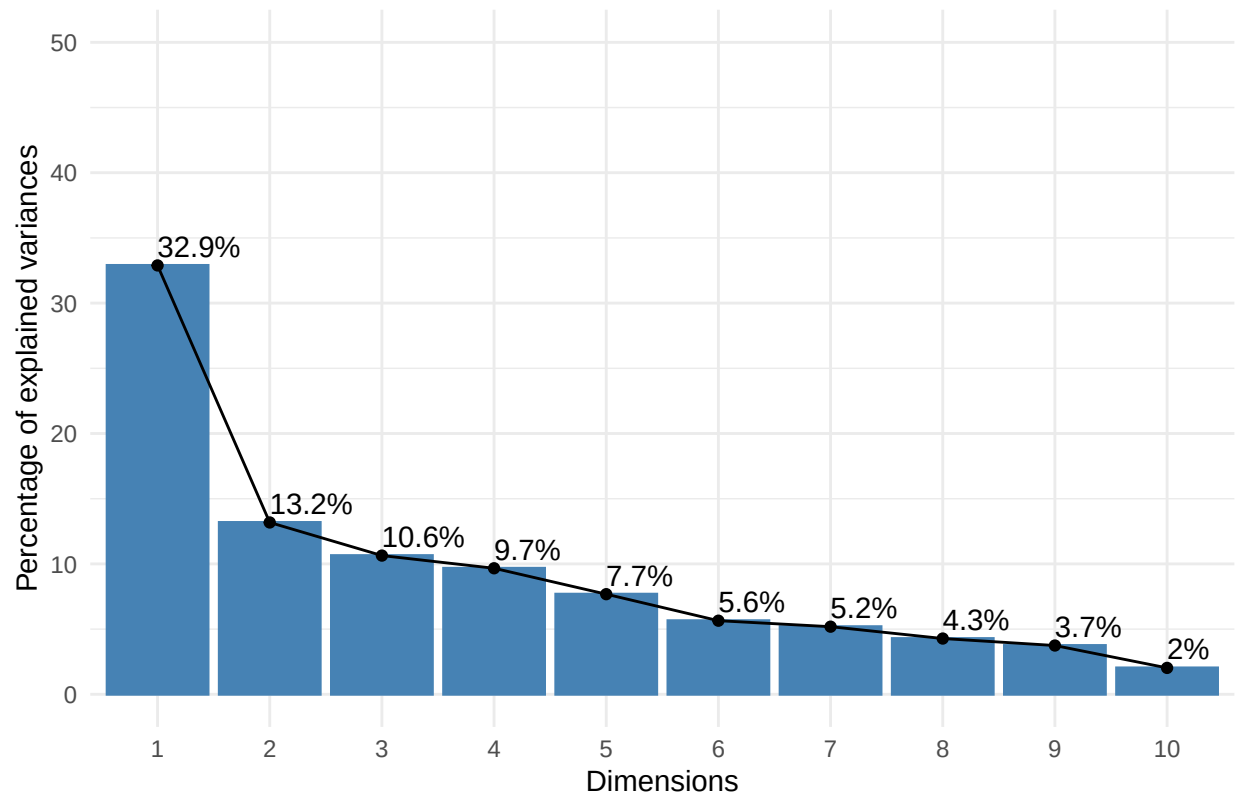


Principal Component Analysis (PCA)

In this analysis, we performed a Principal Component Analysis (PCA) on a set of standardized (per hectare of arable land) agricultural, environmental, and input-related variables across countries. The variables include several yield measures (e.g., cereal, maize, fruit, pulses, sugar crops), energy consumption per hectare (coal, petrol, natural gas), agricultural water withdrawal per hectare, fertilizer input (NPK, herbicides, insecticides), agricultural production value per hectare, and the share of irrigated cropland.

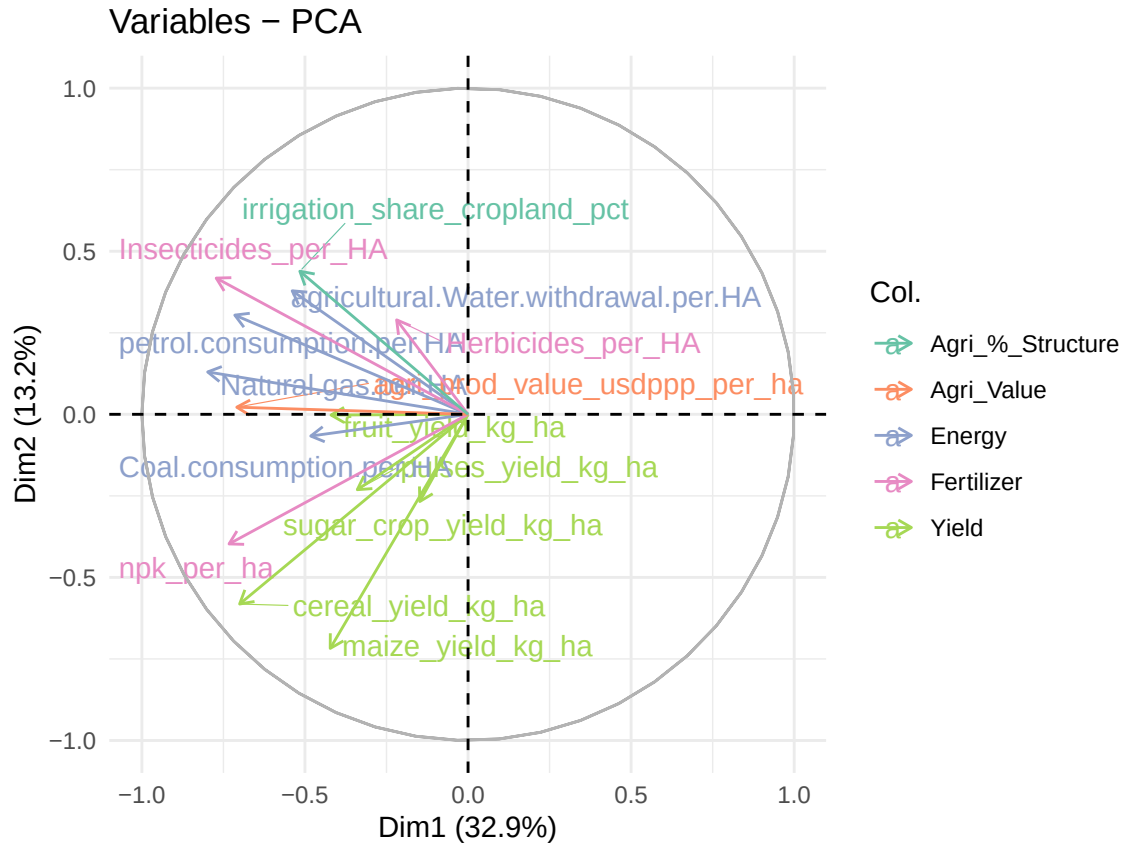
These variables were grouped into thematic categories—Yield, Energy, Fertilizer, Water, Agricultural Structure, Value, and Production—to better understand the multidimensional structure of agricultural efficiency and resource use across countries.

Scree Plot: Variance Explained by Principal Components



The scree plot below shows the percentage of total variance explained by each principal component. The first two principal components explain approximately 33% and 13% of the variance, respectively, accounting for nearly 46% of the total variability.

While the elbow in the scree plot is not very pronounced (with a slight inflection point after the first component), we proceed by focusing on the first two dimensions to explore the main patterns in the data.



This plot shows how each variable contributes to the two main principal components. The direction and length of each arrow represent the contribution and correlation of the variable with the components.

Variables related to agricultural yields cluster in the bottom-left quadrant, suggesting they contribute strongly to Dimension 2. In contrast, variables associated with energy consumption (coal, petrol, gas) are mostly orthogonal to the yield variables and point toward the upper-left quadrant, indicating a strong contribution to Dimension 1. This suggests that yield and energy consumption are not strongly correlated in this dataset.

Similarly, most input-related variables (insecticides, herbicides) align with energy variables, with the exception of NPK fertilizers, which point in the same direction as the yield variables. This may indicate that NPK use is more closely associated with crop productivity than other inputs.

Interestingly, water consumption variables align with energy usage rather than with yields, suggesting that higher water withdrawal does not necessarily translate into higher agricultural performance at a global scale.

The variable representing the average agricultural production value per hectare (USD PPP) points in an intermediate direction. A reasonable interpretation is that energy consumption is more reflective of general development level (e.g., high energy use in oil-exporting countries with low crop yields), rather than of agricultural performance. In contrast, countries with developed agricultural systems and access to irrigation might achieve both higher yield and higher value-added production per hectare. Hence, the average production value captures both productivity and the diversity of economically valuable crops.

Individuals Factor Map (colored by Continent)

